Deliverable 1
Structural Aliasing Analysis in Time-Series Forecasting Using Chronos

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This investigation formalizes the phenomenon of *structural aliasing* within patch-based time-series foundation models, focusing specifically on Amazon's Chronos-Bolt framework.[1, 2] The project focuses on smart-grid load balancing.

Domain Description

To ground this abstract architectural evaluation, the study utilizes a specific physical domain as an empirical reference layer. This reference setting models the solar energy produced by a photovoltaic array attached to an internal load, a local storage battery, and the national electricity grid. The explicit operational objective within this system is to evaluate production variations over short-term future horizons so that the system can dynamically modify the internal load, attach and detach the battery unit, with the aim of maintaining internal grid stability and enabling precise anomaly detection.

Within this infrastructure, a control AI agent is responsible for executing real-time tasks: dynamic load-balancing, battery dispatch, state-monitoring and active anomaly detection, and internal grid stability maintenance.

This analysis is directly intended for researchers, engineers and practitioners utilizing foundation models within the Chronos family for time-series analysis, representation learning, and forecasting, as well as individuals working in the field of energy management who are interested in characterizing the fundamental limitations, structural boundaries, and latent capabilities of these architectures.

Purpose and Scope

This project formalizes and empirically validates the phenomenon of *structural aliasing* in patch-based time-series foundation models, using Amazon Chronos-Bolt-Tiny as the target architecture.

Chronos-Bolt-Tiny operates with a fixed patch size of $P = 16$, stride of $S = 16$ and context window of $T = 2048$ samples as default settings; its architecture comprises four encoder blocks followed by four decoder blocks terminating in a direct quantile projection head.[3, 4]

This study aims to provide a comprehensive understanding of the relationships between raw signal characteristics and the latent representations learned by the model, systematically identifying the critical

During the preparation of this work, the authors used AI-based tools, including GPT-5.5, Gemini 3.6 Flash, Sonnet 5, and Opus 4.6, to support language refinement and improve clarity. All generated or suggested content was carefully reviewed, revised where necessary, and validated by the authors, who assume full responsibility for the final content of this work.

links between sampling frequency (f_s), patch size (P), and stride (S) in order to propose actionable solutions to mitigate the phenomenon.

The core objective is to challenge the baseline assumption that pre-trained temporal foundation models inherently preserve frequency attributes when processing Nyquist-compliant signals.

Mathematical Formulation

The project provides a formal domain description through OWL2 ontology representation of: the physical system (photovoltaic array, battery storage, national grid, internal load), the signal (frequency, amplitude, timespan, phase) and its patched representation (patch size, stride, sampling frequency). See Appendix A for a formal description of the ontology.

The investigation focuses on the formal definition of the phenomenon of structural aliasing, starting from the tokenization operator and describing phase-locking conditions. The project then formalizes the relationship between the signal frequency, patch size, stride, and sampling frequency, providing a mathematical framework to analyze the conditions under which structural aliasing occurs.

Let $\mathbf{x} = \{x[n]\}_{n\in\mathbb{Z}}$ be a discrete-time signal representing the input time series. We define the patch-based tokenization operator $\mathcal{T}_{P,S}$ with patch length $P \in \mathbb{N}^+$ and stride $S \in \mathbb{N}^+$ as a mapping that transforms the signal $\mathbf{x}$ into a sequence of vector-valued tokens $\mathbf{T} = \{\mathbf{t}_k\}_{k\in\mathbb{Z}}$, where each token $\mathbf{t}_k \in \mathbb{R}^P$. The k-th token is defined as

$$\mathbf{t}_k = \mathcal{T}_{P,S}(\mathbf{x})[k] = \begin{bmatrix} x[kS] \\ x[kS+1] \\ \vdots \\ x[kS+P-1] \end{bmatrix}. \tag{1}$$

Consider a continuous-time periodic signal sampled uniformly at sampling frequency f_s, yielding the discrete-time sequence $x[n]$. We isolate a fundamental harmonic component as

$$x[n] = \sin\left(2\pi \frac{f}{f_s} n + \phi\right), \tag{2}$$

where f denotes the signal frequency satisfying the Nyquist criterion $f < \frac{f_s}{2}$, and $\phi \in [0, 2\pi)$ is an arbitrary phase offset.

Phase-locking occurs when the structural phase profile of the signal repeats exactly under a translational shift equal to the stride S. Evaluating a stride translation yields

$$x[n+S] = \sin\left(2\pi \frac{f}{f_s}(n+S) + \phi\right) = \sin\left(2\pi \frac{f}{f_s} n + 2\pi \frac{fS}{f_s} + \phi\right). \tag{3}$$

For exact alignment between the signal and the tokenization operator, the phase accumulation over one stride must correspond to an integer number of 2π–periods

$$2\pi \frac{fS}{f_s} = 2\pi c, \qquad c \in \mathbb{N}^+. \tag{4}$$

As a result, the signal exhibits exact stride-period translational invariance

$$x[n+S] = \sin\left(2\pi \frac{f}{f_s} n + 2\pi c + \phi\right) = \sin\left(2\pi \frac{f}{f_s} n + \phi\right) = x[n]. \tag{5}$$

The same derivation can be applied taking into account translational shifts equal to the patch size P, yielding a similar phase-locking condition

$$2\pi \frac{fP}{f_s} = 2\pi k \Rightarrow x[n+P] = x[n], \qquad k \in \mathbb{N}^+. \tag{6}$$

Solving Equation 4 and Equation 6 for f yields the phase-locking frequency class $\mathcal{F}_{\text{lock}}$, which is defined as the set of frequencies producing stride-period self-repetition

$$\mathcal{F}_{\text{lock}} = \left\{ f \in \mathbb{R}^+ \;\middle|\; f = c\frac{f_s}{S} \vee f = k\frac{f_s}{P},\ c, k \in \mathbb{N}^+,\ f < \frac{f_s}{2} \right\}. \tag{7}$$

The derivation above shows that a stride translation of length S leaves a single stride-locked sinusoid unchanged, $x_f[n+S] = x_f[n]$ for $f \in \mathcal{F}_{lock}$: this repetition is not evidence that two distinct signals become indistinguishable to the model.

Structural aliasing is treated as an emergent, falsifiable property of the trained representations: two frequency-matched but distinct signals x, y structurally alias at layer l if:

$$\|R_l(x) - R_l(y)\|_2 \leq \varepsilon \tag{8}$$

where R_l is the representation at layer l, $\|\cdot\|_2$ is the Euclidean norm, and ε is a pre-registered threshold.

Aliasing implications

The above derivation reframes $\mathcal{F}_{lock}$ as the candidate class where such collisions are hypothesised to occur. In particular, the project tests the following hypotheses.

H1: if $f \in [f_k - \varepsilon, f_k + \varepsilon]$, where $f_k \in \mathcal{F}_{lock}$, the model recovers less information, we expect: higher MDL codelength and lower forecast amplitude recovery than at fixed controls $f_k \pm \delta$, $\delta \gg \varepsilon$. Absent or non-localized loss refutes H1.

H2: the magnitude of degradation at a locked frequency $f_k \in \mathcal{F}_{lock}$ is a property of the stride/patch alignment, not of the signal's own phase: across the S_{f_k} phase offsets already sampled at f_k, the accuracy drop relative to the $f_k \pm \delta$ control is expected to stay approximately constant. A drop whose size varies systematically with phase, rather than only with frequency, refutes H2.

H3: when S is varied while P is fixed, the locations of any local degradation move proportionally to $1/S$ or $1/P$, tracking $\mathcal{F}_{lock}$. An effect that remains tied to P rather than S refutes the proposed stride-lock mechanism and H3.

Data and Evaluation

Two sets of synthetic signals are used to evaluate the presence of structural aliasing in the new models. These include:

- **Sine Wave Signals:** simple sine waves with known frequency systematically varied between 2 Hz and 250 Hz with 1 Hz increments, to assess the model's ability to capture and represent periodic components across a range of frequencies.
- **Time Series Mixup Signals:** 100 signals generated by a sinusoidal mixer based on the TSMixup procedure[1], mixing multiple sine waves of different frequencies; to evaluate the model's capacity to disentangle and represent complex frequency interactions.

This set of 349 test signals is used for all the re-trained models, to evaluate the impact of stride and patch size on the model's ability to capture and represent frequency information.

Optionally, if previous analyses show evidence of aliasing, the project may analyze test signals from a third set, including 100 signals generated using the KernelSynth[1] tool, to test whether the phenomenon persists in more complex signals. The project then focuses on the analysis of real-world data, through the use of the

Run	Patch (P)	Stride (S)	overlap	#patch @2048	steps	time
1	16	8	0.50	255	100k	2.6 h
2	16	12	0.25	170	100k	2.8 h
3	16	16	0.00	128	100k	pending
4	8	8	0.00	256	100k	2.7 h
5	24	24	0.00	86	100k	2.8 h

Table 1: *Patch/stride configurations retrained from scratch (seed 42) under a uniform* 100*k-step budget;* time *is the wall-clock on the RTX 5060 Laptop GPU. The P=16, S=16 baseline is retrained under the* same *budget as the variants, so the stride comparison is not confounded by training budget. The official Amazon checkpoint (*200*k steps, full corpus) is used separately, only as an external forecast-quality reference (Figure 1).*

`Dataset-SolarTechLab.csv` dataset, which contains a one-minute-resolution multivariate telemetry stream of a photovoltaic array[5].

Also, those tests are undertaken for all the re-trained models, to evaluate the impact of stride and patch size on the model's ability to capture and represent frequency information in more complex signals. During the analysis a seed is set to ensure reproducibility of the results.

All signals considered are sampled at $f_s = 512\,\text{Hz}$.

Methodology

Initially, the Chronos-Bolt-Tiny model available on the Hugging Face Hub is used to evaluate the presence of structural aliasing in the default configuration, with patch size $P = 16$ and stride $S = 16$. To evaluate the presence and impact of structural aliasing, the project analyzes a set of synthetic signals with varying frequency characteristics. Then the Chronos-Bolt-Tiny model is tested on simple sine waves with known frequency systematically varied between 2 Hz and 250 Hz with 1 Hz increments, those tests are used to evaluate the presence of structural aliasing.

Since any change to P or S requires retraining, we retrain a restricted grid around the default $P = 16$, $S = 16$

Parameter	Adopted	Chronos official	Max / constraint
Architecture (fixed, from Bolt-tiny)			
`context_length`	2048	2048	Bolt-tiny
`prediction_length`	64	64	Bolt-tiny
quantile levels	9	9	Bolt-tiny
Optimisation			
`batch_size`	32	32	VRAM 8 GB
`max_steps`	100 000	200 000	compute budget
`learning_rate`	10^{-3}	10^{-3}	—
scheduler / warmup	linear / 0	linear / 0	—
`weight_decay`	0.0	0.0	—
`grad_clip_norm`	1.0	1.0	—
optimiser / precision	AdamW (fused) / fp32+TF32	—	
Data pipeline			
corpora (streaming)	TSMixup 10M + KernelSynth 1M		official
mixing ratio	9:1	9:1	—
`shuffle_buffer`	10 000	100 000	streaming timeout
`min_past` / `max_missing` / `drop_prob`	60 / 0.9 / 0.2	—	

Table 2: *Training hyperparameters: adopted value, official Chronos recipe (`chronos-t5-tiny`), and usable maximum / constraint. Only (P, S), the step budget (*100*k vs.* 200*k) and the shuffle buffer (*10*k vs.* 100*k) depart from the official values, to fit the single-GPU budget.*

baseline (see Table 1): two reduced strides at fixed $P = 16$ isolate the effect of patch overlap, while two contiguous patch sizes isolate that of patch width. We exclude $S = 4$: its stride-lock class $\mathcal{F}_{\text{lock}} = \{128, 256, \dots\}$ has one member inside the valid-forecast band; therefore yields no informative local $H1/H3$ contrast on the integer-Hz grid.

Each (P, S) configuration is retrained from scratch: changing P resizes the first patch-embedding block and changing S affects the learned inter-patch dynamics, so weights are not transferable. Loading only the `chronos-bolt-tiny` architecture, the loop follows the *published Chronos-T5* training recipe (fused AdamW, linear LR 10^{-3}, `fp32+TF32`, masked quantile loss) on the two official corpora, TSMixup (10M) and KernelSynth (1M) streamed at the 9:1 ratio[6]; the complete Chronos-Bolt recipe is not publicly released, so this is a budget-matched adaptation of the published hyperparameters rather than an exact reproduction. Every hyperparameter stays at its published value (see Table 2); only (P, S) varies, and a uniform 100k-step budget (vs. 200k) is applied to *all* configurations, including the $P{=}S{=}16$ baseline, so the comparison is not confounded by training budget. A single seed (42) is used for now; multiple seeds remain a planned extension.

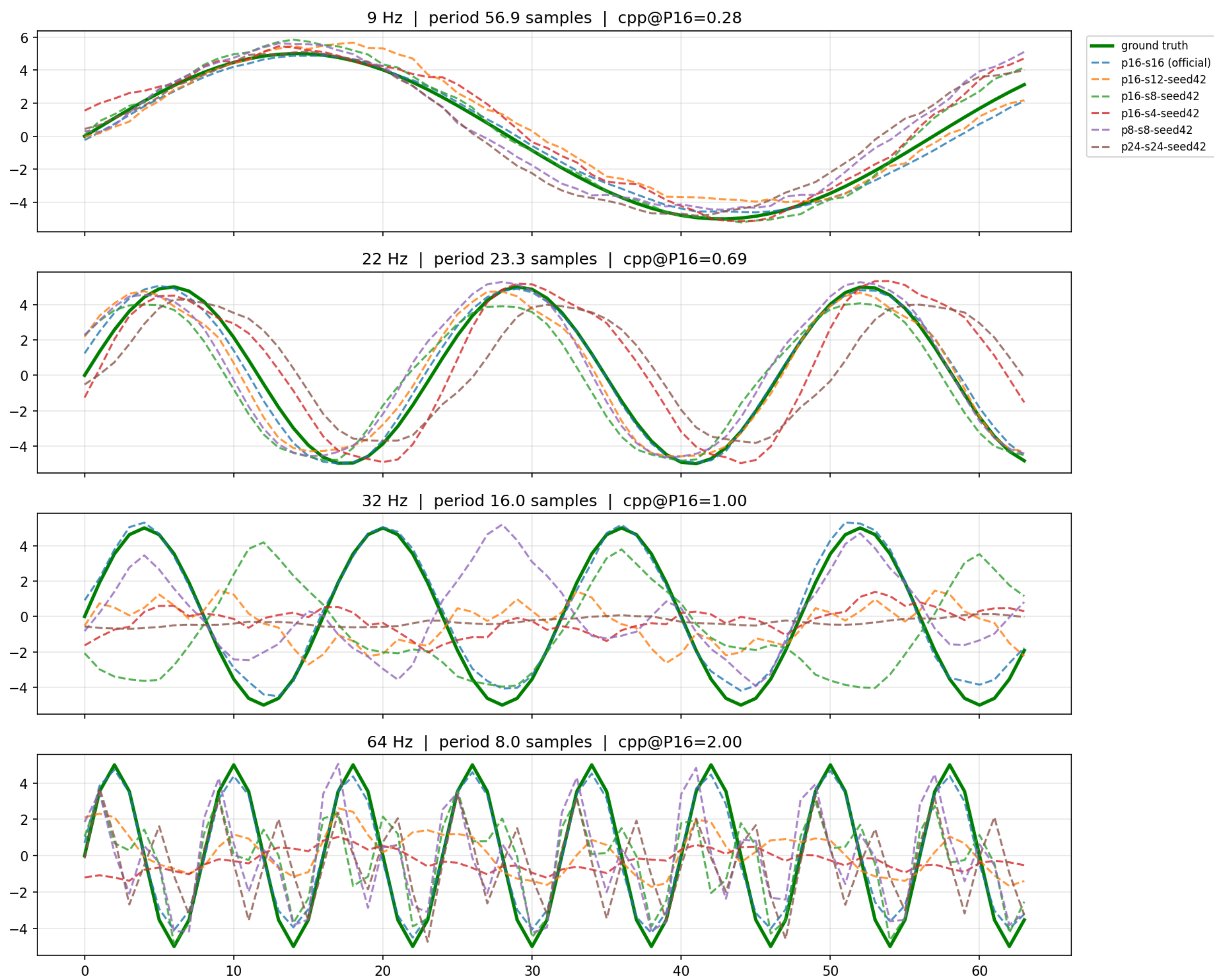

Figure 1: *Forecast quality of the retrained models. Per frequency (rows), each 64-step forecast (dashed) vs. ground truth (green); titles give cycles-per-patch at $P = 16$ (cpp). The real model performances are better than the retrained, due to the larger training budget and more extensive corpora.*

Training ran on one Alienware 16 Aurora (NVIDIA RTX 5060 Laptop GPU, 8 GB VRAM; 32 GB RAM), ≈11 h in total.

Forecasting quality of the retrained models is shown in Figure 1. If necessary, the project may extend the training budget to the full parameters.

Probing Methodology

Following MDL probing[7], the 256-dimensional [REG] representation is captured after each encoder block and each decoder block, plus two points at the output stage: the pre-projection 256-d [REG] input, and the projected quantile output (Pagani et al.'s h_4[8]), where the compression drop reported in the source paper is concentrated. Probing this single per-block vector, rather than the full flattened patch sequence, avoids a dimensionality mismatch across stages and patch/stride variants.

All probe points are compared on the same footing via a prequential-MDL codelength $L(D)$, reported as compression $\mathrm{SV} = 1 - L(D)/L_{\mathrm{uniform}}(D)$, computed independently per probe point for each of the seven hierarchical band-classification tasks[8]. Each value is benchmarked against a shuffled-label control ($\mathrm{SV} \leq 0$ confirms the probe is not memorising noise) and a random-initialisation control (identical untrained architecture, isolating learned information from architectural artefacts).

Independently, a dense per-frequency classification-accuracy profile is obtained at the projected output by sweeping the same band-classification probe across the operational spectrum, with non-redundant phase sampling $S_f = \min\{f_s/\gcd(f, f_s) - 1,\, N\}$ replacing independently drawn random phases.

Bayesian Analysis

We employ a two-part Bayesian framework to quantify both relative local degradation and absolute performance losses across hypotheses $H1$–$H3$. The analysis is conducted only in the frequency range over which valid forecasts are observed, which we expect to be a subset of $[2, 250]$ Hz.

Local Contrast Analysis: to test localized recovery drop at stride-lock for each retrained model configuration (P, S) and training seed, signals at exact $f_k \in \mathcal{F}_{\mathrm{lock}}$ are paired with control signals at offset frequencies $f_k \pm 0.25 f_s/S$, keeping phase, baseline amplitude, noise level, and background process parameters identical. Letting R denote the forecast amplitude recovery ratio, we define the paired logarithmic local contrast d as:

$$d = \log(R_k + 0.01) - \frac{\log(R_- + 0.01) + \log(R_+ + 0.01)}{2} \tag{9}$$

where R_k is the recovery at f_k, and $R_\pm$ are recoveries at the control frequencies $f_k \pm 0.25 f_s/S$. A contrast value of $d < 0$ directly indicates localized forecast attenuation induced by phase-locking.

Contrasts are modeled using a heavy-tailed Student-$t_4(\mu_i, \sigma^2)$ likelihood. The population phase-lock effect $\bar{\beta}$ receives a weakly informative prior $\bar{\beta} \sim$ Student-$t_4(0, 0.5)$, centered at zero due to the absence of prior quantitative estimates in the literature.

Absolute Performance Analysis: to evaluate raw non-negative metric specifically prequential MDL codelength $L(D)$, we use a Bayesian Gamma regression model with a log link:

$$y_i \sim \mathrm{Gamma}(\mathrm{shape} = k,\, \mathrm{mean} = \mu_i), \qquad log(\mu_i) = \alpha_0 + \theta_{\mathrm{lock}} \cdot \mathrm{IsLocked} \tag{10}$$

where $\mathrm{IsLocked} \in \{0, 1\}$ flags phase-lock conditions, and $\theta_{\mathrm{lock}} \sim \mathcal{N}(0, 0.5)$ captures the relative codelength expansion.

The Gamma shape parameter receives a weakly informative prior, $k \sim \mathrm{Gamma}(2, 0.1)$, allowing the model to adaptively capture metric dispersion.

Inference: planned posterior inference will evaluate 95% credible intervals and the posterior probability of at least 20% attenuation.

Security Implications and Risk Assessment

Finally, the project includes a formal risk assessment grounded in the NIST AI Risk Management Framework (AI RMF)[9] to evaluate structural aliasing as a systematic vulnerability within cyber-physical energy systems. This analysis outlines plausible threat models and attack surfaces, possible mitigation strategies (including stride reduction), and an evaluation of whether the proposed mitigation strategies effectively reduce residual exposure for the automated control AI agent.

The risk assessment is structured around the following key points: context and attack surface, threat modeling, impact, mitigation and residual exposure.

Significance and Impact

This subject is relevant because it explores an unexamined systemic vulnerability native to patch-based temporal foundation architectures. By systematically characterizing the conditions under which structural aliasing emerges, this work provides critical insights into the fundamental limitations of these models, directly informing the design of more robust architectures and training paradigms.

For professionals operating within this domain, the analysis could expose how algorithmic data-deletion anomalies can propagate through automated distribution pipelines. By evaluating how hidden token-space representation collapse can deceive a load-balancing AI agent into executing misaligned control choices, this work could provide insights needed for high-reliability, fault-tolerant smart-grid deployments.

The impact of this project could extend beyond theoretical understanding, as it has direct implications for the reliability and safety of real-world applications that depend on accurate time-series forecasting and representation learning, particularly in high-stakes domains such as energy management.

Appendix A: OWL2 Ontology

The core ontology architecture bridges the gap between cyber-physical energy systems and deep learning signal processing domain knowledge. It categorizes the application setting into five main conceptual pillars rooted under the ontological hierarchy:

- **Physical Subsystem (`pv:PhysicalComponent`):** represents the hardware infrastructure of the microgrid, including the solar array, electrochemical storage, consumer loads, and the macro grid tie.
- **Cognitive Layer (`pv:ControlAgent`):** the artificial intelligence system responsible for autonomous grid optimization, load balancing, dispatch scheduling, and proactive vulnerability mitigation.
- **Signal Processing Layer (`pv:Signal`, `pv:PatchedSignal`):** tracks the raw telemetry inputs (both time-domain features and frequency-domain components) along with their tokenized representation configurations (patch dimensions, stride values) tailored for the Chronos foundation architecture.
- **Environmental Tracking (`pv:WeatherObservation`):** captures concurrent meteorological snapshots (air temperature, wind vectors) influencing structural physics and electrical efficiency.
- **Inferred Operational States:** dynamic metadata classified into generation (`pv:GenerationState`), systemic disruptions (`pv:AnomalyState`), control directives (`pv:AgentAction`), and signal degradation hazards (`pv:AliasingEvent`).

Generation States

A formal description of the state of the photovoltaic array is provided through the following axioms, which define the generation state based on the measured tilted irradiance in $\mathrm{W/m^2}$:

$$\begin{gathered}\texttt{PhotovoltaicArray}(?pv) \land \texttt{measures}(?pv, ?s) \land \texttt{tiltedIrradianceWm2}(?s, ?g) \land (?g \geq 800.0) \\ \longrightarrow \texttt{hasGenerationState}(?pv, \texttt{ClearSky})\end{gathered} \tag{11}$$

$$\begin{aligned}\texttt{PhotovoltaicArray}(?pv) \land{} & \texttt{measures}(?pv, ?s) \land \texttt{tiltedIrradianceWm2}(?s, ?g) \\ & \land (?g \geq 150.0) \land (?g < 800.0) \\ & \longrightarrow \texttt{hasGenerationState}(?pv, \texttt{PartialCloud})\end{aligned} \tag{12}$$

$$\begin{aligned}\texttt{PhotovoltaicArray}(?pv) \land{} & \texttt{measures}(?pv, ?s) \land \texttt{tiltedIrradianceWm2}(?s, ?g) \\ & \land (?g > 0.0) \land (?g < 150.0) \\ & \longrightarrow \texttt{hasGenerationState}(?pv, \texttt{Overcast})\end{aligned} \tag{13}$$

$$\begin{gathered}\texttt{PhotovoltaicArray}(?pv) \land \texttt{measures}(?pv, ?s) \land \texttt{tiltedIrradianceWm2}(?s, ?g) \land (?g \leq 0.0) \\ \longrightarrow \texttt{hasGenerationState}(?pv, \texttt{Nighttime})\end{gathered} \tag{14}$$

Anomaly States

Formal description of the anomaly states which could be detected in the photovoltaic array, based on the measured irradiance and the measured power.

Aliasing Event

Formal description of the aliasing event, detected when the frequency of the signal is part of the frequency class $\mathcal{F}_{\text{lock}}$.

Agent Actions

Formal description of the action taken by the control agent, based on the current state and aliasing event.

Appendix B: Changelog

This appendix documents the revisions made to Deliverable 1 in response to the feedback received on the first submission. Entries are grouped under the headings used in that feedback so that each comment can be traced to the corresponding change. The revised document is written under the extended allowance of 2 000 words granted with the feedback.

Feedback point	**Change made**	**Where**
Domain Description before Purpose and Scope	Section order inverted.	Domain Description
Intended Audience shortened or integrated	Standalone section removed; content condensed into a single paragraph.	Domain Description, final paragraph
Significance and Impact after the project outline	Moved from third to last section of the body.	Significance and Impact
Dedicated Data section	Added; test signals and datasets no longer described across several sections.	Data and Evaluation
State the intended mathematical formulation	New section: tokenisation operator, phase-locking conditions, locked frequency class, operational definition of structural aliasing.	Mathematical Formulation, Eqs. (1)–(8)
Reformulate the hypotheses explicitly	Objectives replaced by H1–H3, each with a predicted direction, the metric used and an explicit refutation criterion.	Aliasing implications
Make the probing experiments precise	New section stating the probed representations, the metric, the probe points compared and the controls.	Probing Methodology
Simplify the experimental pipeline	Staged as: default checkpoint on clean sinusoids, then the retrained (P, S) grid, then complex synthetic signals, then the photovoltaic data.	Data and Evaluation; Methodology
Clarify whether stride changes require retraining	Retraining shown to be necessary for both P and S; the training recipe, its departures from the published one and their reasons are tabulated.	Methodology, Tables 1–2
Explore strides on both sides of the default	Admissibility constraint stated and the symmetric design built within it.	Methodology
Describe the planned priors	Likelihoods, link function and all priors stated, with justification.	Bayesian Analysis, Eqs. (9)–(10)

Table 3: *Summary of the feedback received on the first submission and of the corresponding revisions.*

Structure and organisation

- The order of the opening sections was inverted so that the application context precedes the statement of purpose. The document now opens with Domain Description, followed by Purpose and Scope.
- The standalone *Intended Audience* section, previously two paragraphs, was removed. Its content was condensed into a single paragraph closing the Domain Description.
- *Significance and Impact* was moved to the end of the body, after the reader has seen the formalisation, the data, the experimental design and the risk assessment.
- A dedicated *Data and Evaluation* section was added. It now holds the synthetic sine sweep, the sinusoidal-mixer set, the KernelSynth contingency and the photovoltaic dataset, which in the first submission were split between *Purpose and Scope* and *Project Outline*.
- The single *Project Outline* section was replaced by separate sections for the formalisation, the data, the training procedure, the probing protocol, the statistical analysis and the risk assessment.

- An appendix was added for the domain formalisation, so that the ontology does not consume the word allowance of the body.

Mathematical formulation and hypotheses

- A *Mathematical Formulation* section was added. The first submission announced the formalisation as a planned activity but did not outline it.
- The patch-based tokenisation operator $\mathcal{T}_{P,S}$ is defined explicitly as a map from the sampled signal to a sequence of vector-valued tokens, Eq. (1).
- The signal model and the Nyquist constraint under which the phenomenon is studied are stated, Eq. (2).
- The stride-translation derivation is given in full: the phase accumulated over one stride, Eq. (3), the alignment condition on that accumulation, Eq. (4), and the resulting exact stride-period invariance, Eq. (5). The analogous condition for a shift of one patch length follows as Eq. (6).
- The two conditions are collected into the phase-locking frequency class $\mathcal{F}_{\text{lock}}$, Eq. (7), expressed as a function of f_s, P and S.
- It is now stated explicitly that self-repetition of consecutive patches is a property of a single signal and is not by itself evidence that two distinct signals become indistinguishable. The derived invariance and the empirical claim are therefore kept separate, and structural aliasing is given an operational, falsifiable definition at the level of the learned representations, Eq. (8).
- The three items previously listed as objectives were reformulated as hypotheses H1, H2 and H3. Each now states a predicted direction, the quantity in which the effect should appear, the comparison against which it is measured, and the observation that would refute it.
- The ontology, previously a single bullet, is set out in Appendix A. The generation-state axioms are given in full; the anomaly, aliasing-event and agent-action axioms are described but not yet formalised, and are submitted at this stage for feedback on the modelling approach before the remaining rules are written.

Experimental methodology

- A *Probing Methodology* section was added, replacing the single sentence of the first submission.
- The probed representation is now specified: the 256-dimensional [REG] vector captured after each encoder block and each decoder block, together with the pre-projection input and the projected quantile output. The reason for probing this per-block vector rather than the flattened patch sequence is given.
- The first submission probed only after each decoder block. Probing the encoder as well allows the stage at which frequency information is lost to be located rather than merely detected.
- The metric is specified as a prequential MDL codelength, reported as a normalised codelength saving so that probe points of different dimensionality are compared on the same footing, together with the classification tasks on which it is computed and two controls, one with shuffled labels and one with an untrained network of identical architecture.
- A dense per-frequency accuracy profile was added, with deterministic non-redundant phase sampling in place of independently drawn random phases.
- The pipeline was simplified and ordered: the theoretical predictions are first tested on clean synthetic sinusoids with the released checkpoint, the architectural blind spots are then localised across the retrained configurations, and the photovoltaic dataset is used last, as an application case.

Patch stride experiments

- It is now stated that any change to P or S requires retraining from scratch, with the reason for each: a change of P resizes the patch-embedding block, and a change of S alters the inter-patch dynamics the model has learned. The first submission announced a sweep over $S \in [8, 16]$ without addressing this point.

- Table 1 reports the configurations retrained for this submission, with overlap, number of patches over the context window, step budget and wall-clock time.
- Table 2 compares every training hyperparameter against the published Chronos recipe and isolates the only departures from it, each with the constraint that motivates it.
- It is stated that the complete Chronos-Bolt training recipe is not publicly released, so the training loop is a budget-matched adaptation of the published Chronos-T5 recipe rather than an exact reproduction. The reduced step budget is applied uniformly to every configuration, including the default one, so that comparisons across stride values are not confounded by training budget. The released checkpoint is used only as an external forecast-quality reference.
- The exclusion of $S = 4$ is justified rather than left implicit.
- The context window of Chronos-Bolt-Tiny is corrected to $T = 2048$ samples; the first submission reported $T = 512$ tokens.
- Preliminary training runs are reported, with the hardware used and the total wall-clock time. Figure 1 is included as evidence that the proposed configurations train to a usable state within the available budget; it is not offered as an analysis of the phenomenon, which belongs to the later deliverables.

Bayesian analysis

- A *Bayesian Analysis* section was added, replacing the single sentence of the first submission.
- The analysis is now split into a relative and an absolute part. The relative part pairs each locked frequency with matched controls either side of it and reduces the pair to a logarithmic contrast, Eq. (9), so that the localised effect is separated from the overall trend in forecast quality with frequency.
- The likelihood for the contrasts is stated, and a heavy-tailed choice is adopted to limit the influence of individual configurations. The prior on the population effect is given together with the reason for centring it at zero.
- The absolute part is specified as a Gamma regression with a log link, Eq. (10), with the priors on the lock coefficient and on the dispersion parameter stated explicitly.
- The planned inference is stated: credible intervals and the posterior probability of an effect exceeding a pre-registered size, rather than point estimates alone.

References

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